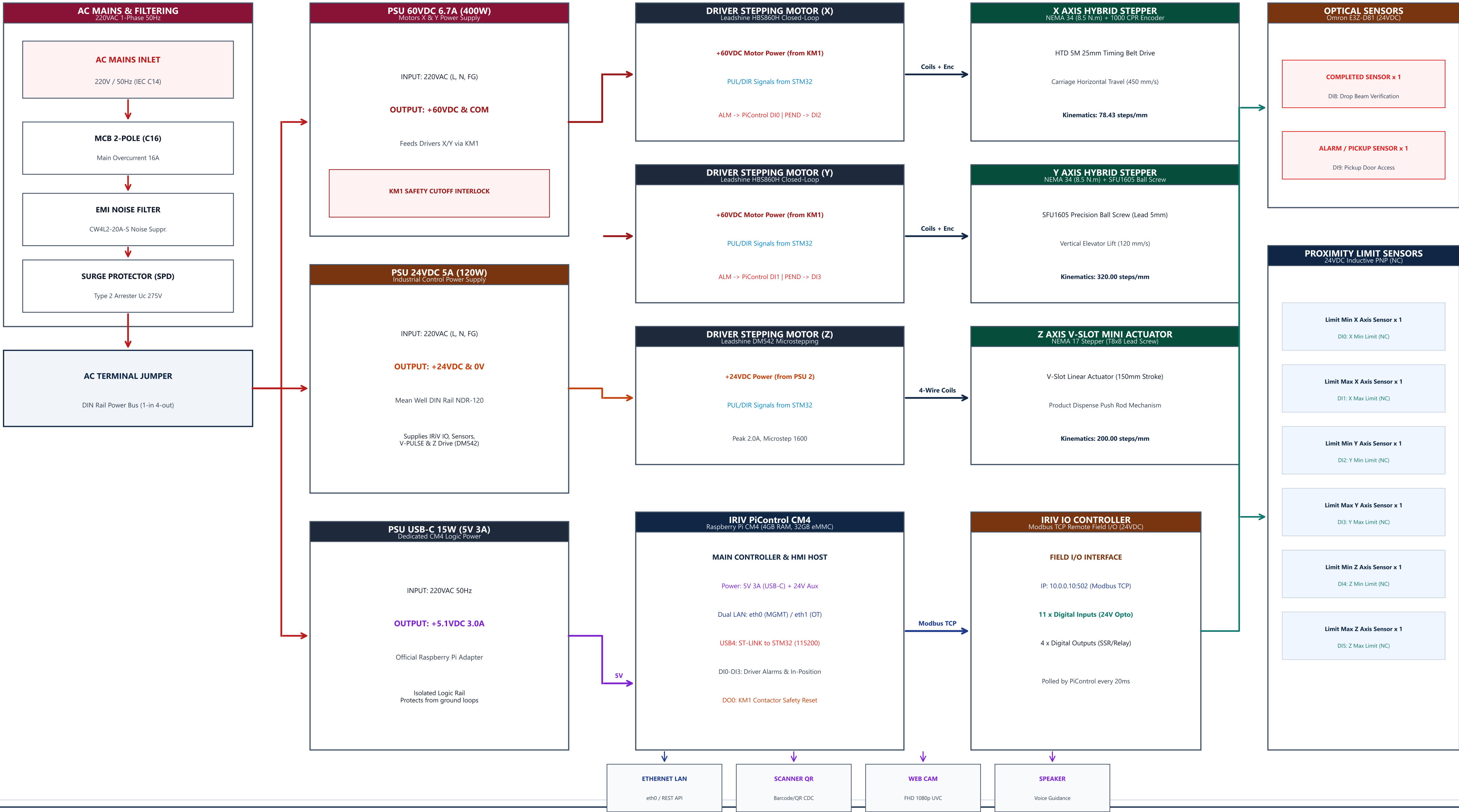


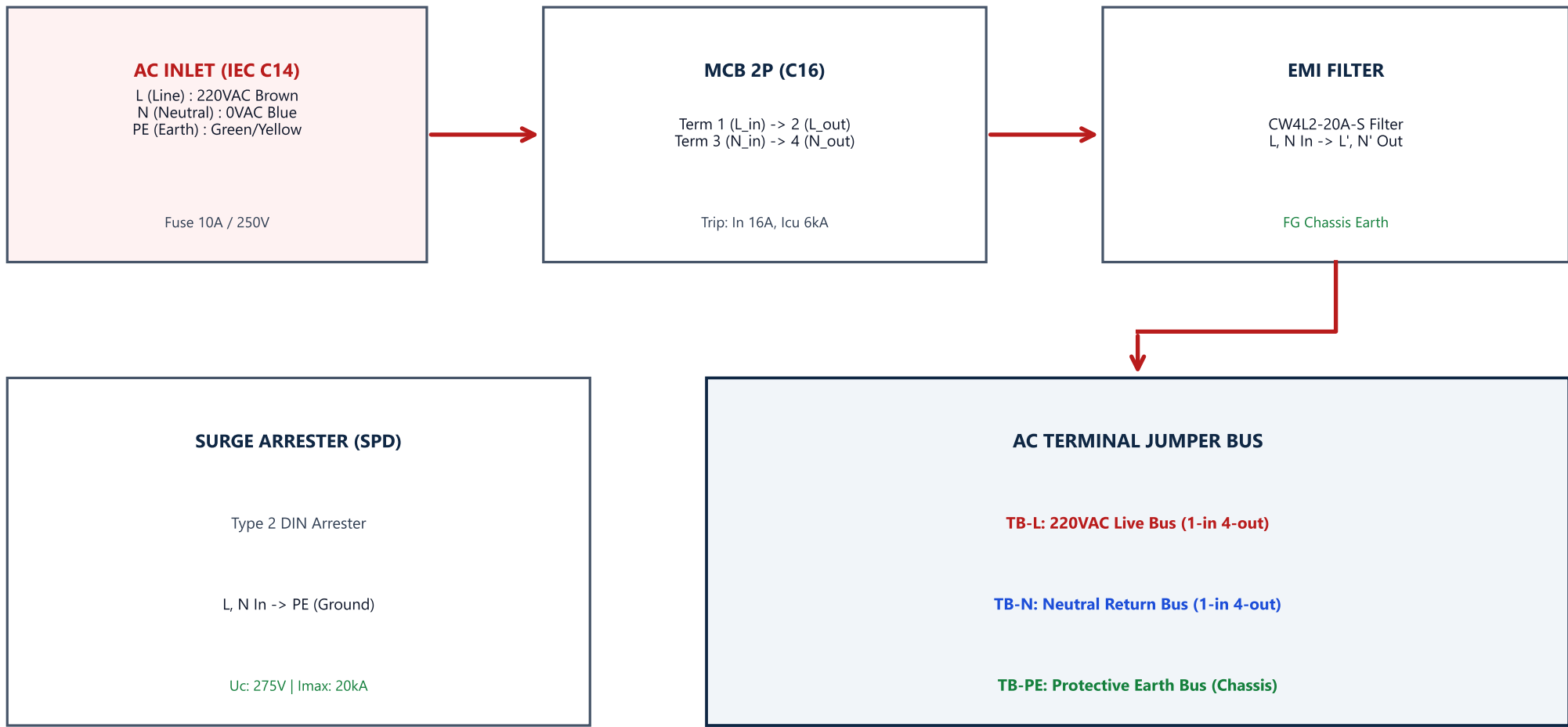
PROJECT: NARIT SMART VENDING MACHINE | SYSTEM ARCHITECTURE & ELECTRICAL FLOW BLOCK DIAGRAM

SECTION: OVERVIEW | VOLTAGE: 220VAC / 60VDC / 24VDC / 5VDC



AC MAINS & POWER INTERLOCK SCHEMATIC

Single-Phase 220VAC 50Hz, Circuit Protection, and KM1 Safety Contactor



KM1 SAFETY CONTACTOR & HARDWARE POWER CUTOFF

MAIN CONTACTS (1-2)

+60VDC Line In (PSU 1) -> +60VDC Out to Drivers X/Y

MAIN CONTACTS (3-4)

0VDC Return In (PSU 1) -> 0VDC Out to Drivers X/Y

COIL TERMINALS (A1-A2)

24VDC Coil driven by PiControl DO0 in series with E-Stop NC

AUX CONTACT (21-22 NC)

Hardware Armature Feedback wired to IRIV IO DI10 (Fail-Safe)

AUX CONTACT (13-14 NO)

Spare Auxiliary Signaling / Indicator Light

SAFETY PRINCIPLE: E-Stop hard-drops coil voltage without software dependency.

POWER SUPPLIES TERMINAL PINOUT SCHEDULE

Terminal Numbers, Signal Names, Voltages, Wire Gauges & Color Codes

1. PSU 1: 60VDC 6.7A (400W) TERMINAL PINOUT:

TERM	PIN NAME	VOLTAGE / TYPE	WIRE GAUGE	DESTINATION / REMARKS
L	AC Input Line	220VAC 50Hz	2.5 mm² Brown	From TB-L Bus
N	AC Input Neutral	0VAC Neutral	2.5 mm² Blue	From TB-N Bus
FG	Chassis Earth	Ground Bar	2.5 mm² Grn/Yel	From TB-PE Earth Bar
+V (x2)	+60VDC Output	+60VDC Bus	2.5 mm² Red	To KM1 Contact 1 (L1)
-V (x2)	0VDC Output	0V Return / COM	2.5 mm² Black	To KM1 Contact 3 (L2)
V-ADJ	Voltage Trimmer	58V - 62V Adj.	Internal Pot	Calibrated to 60.0VDC

2. PSU 2: 24VDC 5A (120W MEAN WELL) TERMINAL PINOUT:

TERM	PIN NAME	VOLTAGE / TYPE	WIRE GAUGE	DESTINATION / REMARKS
L	AC Input Line	220VAC 50Hz	1.5 mm² Brown	From TB-L Bus
N	AC Input Neutral	0VAC Neutral	1.5 mm² Blue	From TB-N Bus
FG	Chassis Earth	Ground Bar	1.5 mm² Grn/Yel	From TB-PE Earth Bar
+24V (x2)	+24VDC Output	+24VDC Control	1.5 mm² Orange	IRIV IO, Sensors, V-PULSE
0V (x2)	0VDC Output	0V Return / COM	1.5 mm² Black	IRIV IO, Sensors 0V, MCU GND
DC OK	Relay Contact	Dry Contact	0.5 mm² Pair	Optional PSU status alert

3. PSU 3: 5VDC 3A (15W USB-C) TERMINAL PINOUT:

TERM	PIN NAME	VOLTAGE / TYPE	WIRE GAUGE	DESTINATION / REMARKS
AC IN	2-Pin Euro/US	100-240VAC	Molded Cord	From AC Socket / Bus
VBUS	USB-C Pin A4/B9	+5.1VDC Power	18 AWG Wire	PiControl USB-C Port
GND	USB-C Pin A1/B12	0V Power Return	18 AWG Wire	PiControl USB-C Port
ISOLATION	Galvanic Barrier	Reinforced 3kV	Internal Xfmr	Prevents Ground Loops

ELECTRICAL CODE & WIRE COLOR SPECIFICATIONS (IEC 60204-1):

- AC Mains Power Lines: Phase = Brown (2.5 mm²), Neutral = Light Blue (2.5 mm²), Earth = Green/Yellow.
- DC High Voltage Power (+60V): Red (2.5 mm²), Return COM = Black (2.5 mm²). Separated from logic.
- DC Low Voltage Control (+24V): Orange (1.0 mm²), Return (0V) = Black (1.0 mm²).
- Earth Bonding: Cabinet metal frame, din rails, and motor casings bonded to PE bar with <= 0.1 Ohm.
- Wireways: Power wires (AC 220V, DC 60V) routed in left duct; low-voltage signals in right duct.

Cytron IRiV PiControl CM4 (MAIN CONTROLLER)

Raspberry Pi CM4 (4GB RAM, 32GB eMMC, Dual Ethernet, Isolated GPIOs)

1. DUAL GIGABIT ETHERNET & USB PORTS:

INTERFACE	IP / PROTOCOL	CONNECTOR	FUNCTION & DESTINATION
eth0 (MGMT)	192.168.70.80 / 24	RJ45 (GbE)	HMI Kiosk Web Port 80, REST API, MQTT Broker
eth1 (OT LAN)	10.0.0.2 / 24	RJ45 (GbE)	Modbus TCP Master to IRiV IO (10.0.0.10:502)
USB 1	USB CDC Serial	USB 2.0 Type-A	2D Barcode & QR Code Payment Scanner
USB 2	USB Video UVC	USB 2.0 Type-A	FHD 1080p Web Camera (Drop / Pickup Vision)
USB 3 / 3.5mm	Audio DAC Output	3.5mm / USB	Voice Guidance & Chime Speaker (3W)
USB 4	115200 8-N-1 VCP	USB 2.0 Type-A	Safe-Link v3 Binary Protocol to STM32 NUCLEO

2. LOCAL ISOLATED GPIO TERMINAL PINOUT (24VDC CHANNELS):

TERMINAL	BCM PIN	SIGNAL NAME	DIRECTION	ELECTRICAL FUNCTION & INTERLOCK
DI0	GPIO 13	X_DRIVE_ALM	Input (Opto)	Driver X Alarm (Active HIGH, Fail-safe)
DI1	GPIO 17	Y_DRIVE_ALM	Input (Opto)	Driver Y Alarm (Active HIGH, Fail-safe)
DI2	GPIO 27	X_PEND	Input (Opto)	Driver X In-Position Verification Signal
DI3	GPIO 22	Y_PEND	Input (Opto)	Driver Y In-Position Verification Signal
DO0	GPIO 23	XY_DRIVE_POWER	Output (Relay)	KM1 Safety Contactor Driver (3.0s Auto-Reset)
DO1	GPIO 24	RELAY_AUX_1	Output (Relay)	Auxiliary 24V SSR Driver (Spare Channel 1)
DO2	GPIO 25	RELAY_AUX_2	Output (Relay)	Auxiliary 24V SSR Driver (Spare Channel 2)
DO3	GPIO 16	RELAY_AUX_3	Output (Relay)	Auxiliary 24V SSR Driver (Spare Channel 3)
24V IN	Power V+	+24VDC Supply	Power In	From PSU 2 Control Rail (Orange wire)
0V IN	Power V-	0VDC Return	Power In	From PSU 2 Control Rail (Black wire)

SYSTEM SOFTWARE SERVICES & RUNTIME DAEMONS:

- narit-vending-web-iriv.service : Flask Touchscreen Kiosk UI on localhost:80.
- narit-vending-controller-iriv.service : Core safety state machine and hardware gateway.
- IPC Socket Bus : /run/narit-vending/ctrl.sock (Non-blocking Unix Domain Socket).
- Cyclic Modbus Polling : Master polls IRiV IO every 20ms; Disarms if data stale > 350ms.
- Safe-Link v3 Motion Protocol : CRC-16 validated frames; 500ms heartbeat timeout watchdog.
- Hardware Fault Reset : PiControl DO0 drops KM1 for 3s to discharge motor driver fault registers.
- Telemetry Cloud Sync : MQTT client connects to broker.emqx.io:1883 for remote monitoring.
- Database : Local SQLite database persists inventory slots, motor kinematics & logs.

Cytron IRiV IO REMOTE FIELD CONTROLLER

Modbus TCP Slave (IP: 10.0.0.10:502, Unit ID: 255) — 11 DI, 4 DO

1. DIGITAL INPUTS (DI0 - DI10) PINOUT & MAPPING:

CH	NAME	CONNECTED SENSOR	LOGIC STATE	BEHAVIOR & SAFETY ACTION
DI0	x_head_limit	Limit Min X Axis Sensor	Normally Closed (NC)	Stops X Reverse Motion (Home)
DI1	x_tail_limit	Limit Max X Axis Sensor	Normally Closed (NC)	Stops X Forward Motion
DI2	y_head_limit	Limit Min Y Axis Sensor	Normally Closed (NC)	Stops Y Down Motion (Drop level)
DI3	y_tail_limit	Limit Max Y Axis Sensor	Normally Closed (NC)	Stops Y Up Motion (Top level)
DI4	z_head_limit	Limit Min Z Axis Sensor	Normally Closed (NC)	Stops Z Retract (Pusher Home)
DI5	z_tail_limit	Limit Max Z Axis Sensor	Normally Closed (NC)	Stops Z Extend (Pusher Max)
DI6	z_home	Z Home Reference Sensor	Normally Open (NO)	Reference datum sensor for Z
DI7	drop_park	Drop Alignment Proximity	Normally Open (NO)	Confirms carriage aligns with chute
DI8	drop_sens	Completed Sensor (Omron)	Beam Interrupted	Verifies product successfully dropped
DI9	pick_sens	Alarm Sensor (Omron E3Z)	Door Flap Opened	Detects user hand / door open access
DI10	estop_fb	KM1 Contactor Aux NC	Fail-Safe NC Loop	Active LOW; loop break trips E-Stop

2. DIGITAL OUTPUTS (DO0 - DO3) PINOUT & INDICATORS:

CH	NAME	LOAD DEVICE	OUTPUT TYPE	OPERATIONAL CONDITION
DO0	ready	Green Pilot Light	24V / 0.5A SSR	Solid ON when system armed & idle
DO1	moving	Yellow Pilot Light	24V / 0.5A SSR	Blinks during carriage X/Y movement
DO2	alarm	Red Light + Audible Buzzer	24V / 0.5A SSR	Active during drive fault, E-Stop or stall
DO3	dispense	Solenoid Drop Gate Relay	24V Interposing	500ms pulse triggers pickup door latch

IRiV IO SENSOR TERMINATION & NOISE REJECTION RULES:

- S/S Jumper Rule : The S/S terminal MUST be tied to 0V for PNP sensors (Sinking input circuit).
- Fail-Safe NC Limit Inputs : Limit sensors DI0-DI5 use NC logic. Wire break stops motion immediately.
- Cable Shielding : All sensor cables must use shielded pairs; shield tied to PE ground bar at cabinet.
- Debounce Filters : Hardware optocouplers + 3-sample software digital filter suppresses contact bounce.
- Inductive Load Protection : DO3 solenoid coil requires flyback diode (1N4007) across terminals.
- Stale Guard Policy : Software watchdog resets all DOs to OFF if Modbus link is interrupted > 350ms.
- Modbus Addressing : Unit ID = 255 | Discrete Inputs: 10001-10011 | Coils: 00001-00004.

OPTICAL PHOTOELECTRIC SENSORS (OMRON E3Z-D81)
Product Drop Verification & Door Access Safety Sensors (24VDC Infrared)

1. COMPLETED SENSOR x 1 (OMRON E3Z-D81 / E3Z-R81)

Mounting: Dispense Chute Beam | Function: Confirms product drop onto tray

WIRE COLOR	TERMINAL PIN	VOLTAGE / SIGNAL	CONNECTION AT IRIV IO
BROWN (BN)	Pin 1 (Power +)	+24VDC Supply	From PSU 2 (+24VDC Rail)
BLUE (BU)	Pin 3 (Power -)	0VDC Return	From PSU 2 (0VDC Common Rail)
BLACK (BK)	Pin 4 (Output)	PNP Open Collector	To IRIV IO Terminal DI8 (Drop Sensor)
WHITE (WH)	Pin 2 (Mode)	Dark-ON / Light-ON	Tied to Brown for Dark-ON mode

2. ALARM / PICKUP SENSOR x 1 (OMRON E3Z-D81)

Mounting: Pickup Flap Door | Function: Detects door open / hand access

WIRE COLOR	TERMINAL PIN	VOLTAGE / SIGNAL	CONNECTION AT IRIV IO
BROWN (BN)	Pin 1 (Power +)	+24VDC Supply	From PSU 2 (+24VDC Rail)
BLUE (BU)	Pin 3 (Power -)	0VDC Return	From PSU 2 (0VDC Common Rail)
BLACK (BK)	Pin 4 (Output)	PNP Open Collector	To IRIV IO Terminal DI9 (Pickup Sensor)
WHITE (WH)	Pin 2 (Mode)	Dark-ON / Light-ON	Tied to Brown for Dark-ON mode

OPTICAL SENSOR CALIBRATION PROCEDURE:

1. Sensitivity Trimmer: Turn clockwise until yellow output LED lights with target present.

2. Beam Alignment: Verify infrared beam reflects reliably across full width of dispense chute.

3. Interlock: Carriage motion is locked in HOLD while DI9 (Pickup Door) is asserted.

INDUCTIVE PROXIMITY LIMIT SENSORS (AXES X, Y, Z)
6 x Inductive Proximity Sensors (24VDC PNP Normally Closed)

PROXIMITY LIMIT SENSOR PINOUT & SCHEDULE (DI0 - DI5):

AXIS / LIMIT	SENSOR MODEL	IRIV IO PIN	NORMAL / ACTIVE	MECHANICAL FUNCTION
LIMIT MIN X	LJ12A3-4-Z/AX (PNP)	DI0 (Channel 0)	Closed (NC) -> High	X-Axis Min / Home Limit (Left)
LIMIT MAX X	LJ12A3-4-Z/AX (PNP)	DI1 (Channel 1)	Closed (NC) -> High	X-Axis Max Overtravel (Right)
LIMIT MIN Y	LJ12A3-4-Z/AX (PNP)	DI2 (Channel 2)	Closed (NC) -> High	Y-Axis Bottom Limit (Drop Chute)
LIMIT MAX Y	LJ12A3-4-Z/AX (PNP)	DI3 (Channel 3)	Closed (NC) -> High	Y-Axis Top Overtravel (Level 5)
LIMIT MIN Z	LJ12A3-4-Z/AX (PNP)	DI4 (Channel 4)	Closed (NC) -> High	Z-Axis Retract Home (Pusher Back)
LIMIT MAX Z	LJ12A3-4-Z/AX (PNP)	DI5 (Channel 5)	Closed (NC) -> High	Z-Axis Extend Max (Pusher Forward)
Z HOME REF	LJ12A3-4-Z/BX (PNP)	DI6 (Channel 6)	Open (NO) -> High	Z Secondary Home Datum Sensor
PARKING SENS	LJ12A3-4-Z/BX (PNP)	DI7 (Channel 7)	Open (NO) -> High	Carriage Drop Alignment Sensor

STANDARD 3-WIRE INDUCTIVE PROXIMITY WIRING (PNP NC):

WIRE COLOR	CORE FUNCTION	SUPPLY RAIL	TERMINATION POINT
BROWN (BN)	DC Positive V+	+24VDC Power Bus	PSU 2 (+24VDC Terminal Bus)
BLUE (BU)	DC Common V-	0VDC Power Return	PSU 2 (0VDC Common Bus)
BLACK (BK)	Signal Output	PNP Switching Load	IRIV IO DI0 to DI7 Terminals
SHIELD (DRAIN)	Noise Screen	Earth Ground (PE)	Cabinet PE Ground Bar (One end only)

S/S Terminal on IRIV IO MUST be jumpered to 0V for PNP sensor operation.

Sensing Distance: 4.0 mm for ferrous targets (Steel bracket).

FAIL-SAFE NORMALLY CLOSED (NC) SAFETY PHILOSOPHY:

• Under normal operation, proximity sensors maintain closed circuit (+24V asserted at DIx).

• When carriage reaches mechanical limit, metal flag separates and signal drops to 0V.

• CRITICAL: If a wire breaks or comes loose, signal instantly drops to 0V (Triggers E-Stop).

• Prevents machine run-away that would occur with Normally Open (NO) sensors.

• DI10 (KM1 Aux Feedback) uses identical active-LOW fail-safe loop logic.

DRIVERS X & Y: LEADSHINE HBS860H

Hybrid Closed-Loop Drives for X & Y Axes (Peak 8.2A, +60VDC)

TERMINAL	SIGNAL NAME	CONNECTION & ELECTRICAL SPEC
AC / AC	+60VDC / 0VDC	+60V Power Bus from KM1 Contactor Out
PUL+ (V-PULSE)	+24VDC Common	+24VDC Common Anode Bus (PSU 2)
PUL- (STEP)	Step Pulse In	From NMOS Ch1 (X) / Ch3 (Y) Drain
DIR+ (V-PULSE)	+24VDC Common	+24VDC Common Anode Bus (PSU 2)
DIR- (DIR)	Direction In	From NMOS Ch2 (X) / Ch4 (Y) Drain
ENA+ / ENA-	Enable Input	Not connected (Driver defaults enabled)
ALM+ / ALM-	Alarm Output	Optocoupler to PiControl DI0 (X) / DI1 (Y)
PEND+ / PEND-	In-Position Out	Optocoupler to PiControl DI2 (X) / DI3 (Y)
A+ / A-	Phase A Coils	To 86HBS85 Motor Phase A (Red / Blue)
B+ / B-	Phase B Coils	To 86HBS85 Motor Phase B (Black / Green)
EA+ / EA-	Encoder Ch A	Differential Optical Encoder Channel A
EB+ / EB-	Encoder Ch B	Differential Optical Encoder Channel B
VCC / EGND	Encoder +5V/0V	+5VDC Power to Encoder (Red / White)

DIP SWITCH SETTINGS (HBS860H):

- Current (SW1 - SW4): ON, ON, ON, ON -> Peak 8.2A (5.6A RMS)
- Microstep (SW5 - SW8): ON, OFF, ON, ON -> 1600 pulses/revolution
- Direction (SW9): OFF -> Normal rotation | SW10: OFF -> Closed loop active

MOTOR SPECIFICATIONS (AXES X & Y):

- Model: 86HBS85 Hybrid Closed-Loop Stepper (NEMA 34)
- Holding Torque: 8.5 N.m | Rated Current: 5.6A / phase
- Encoder: 1000 CPR Optical Quadrature (4000 counts/rev)
- Axis X Transmission: HTD 5M 25mm Belt -> 78.43 steps/mm
- Axis Y Transmission: SFU1605 Ball Screw -> 320.00 steps/mm
- Max Speed X: 450 mm/s | Accel X: 800 mm/s² (Dual HGR20)
- Max Speed Y: 120 mm/s | Accel Y: 600 mm/s² (Elevator Lift)
- Stall Protection: Drive trips ALM if encoder lags > 1000 pulses.

DRIVER Z: LEADSHINE DM542

Digital Microstepping Drive for Pusher Axis (Peak 4.2A, +24VDC)

TERMINAL	SIGNAL NAME	CONNECTION & ELECTRICAL SPEC
V+ / GND	+24VDC / 0VDC	+24VDC Control Supply from PSU 2 Rail
PUL+ (V-PULSE)	+24VDC Common	+24VDC Common Anode Bus (PSU 2)
PUL- (STEP)	Step Pulse In	From NMOS Ch5 (PA5 / D13) Drain
DIR+ (V-PULSE)	+24VDC Common	+24VDC Common Anode Bus (PSU 2)
DIR- (DIR)	Direction In	From NMOS Ch6 (PB2) Drain
ENA+ / ENA-	Enable Input	Not connected (Driver defaults enabled)
A+ / A-	Phase A Coils	To NEMA 17 Pusher Phase A (Red / Blue)
B+ / B-	Phase B Coils	To NEMA 17 Pusher Phase B (Black / Green)

DIP SWITCH SETTINGS (DM542):

- Current (SW1 - SW3): ON, OFF, ON -> Peak 2.0A (1.4A RMS)
- Standstill (SW4): OFF -> Half current during standstill (idle)
- Microstep (SW5 - SW8): ON, OFF, ON, ON -> 1600 pulse/rev

MOTOR & ACTUATOR SPECIFICATIONS (AXIS Z):

- Model: NEMA 17 Stepper (42x42mm, 1.5A / phase)
- Actuator: V-Slot Mini Lead Screw Linear Actuator
- Lead Screw: T8x8 (Pitch 2mm, 4 Starts = Lead 8.0mm/rev)
- Stroke Length: 150mm Pusher Rod Travel
- Kinematic Ratio: 200.00 steps/mm @ 1600 microstep
- Push Speed: 50 mm/s | Retract Speed: 80 mm/s

SAFETY INTERLOCK GATES (AXIS Z):

- Carriage X/Y motion is locked unless DI4 (Z_MIN) is asserted.
- Push cycle extends to DI5 (Z_MAX), pauses 500ms, retracts.
- Soft-limit timeout: Auto-reverses if stroke exceeds 3.5s.

- Safety Gap: 24V supply recommended to route via KM1 contactor.

STM32 & 6-CH NMOS PULSE INTERFACE

Hardware Timer Output Compare Pulse Engine + Low-Side Sink Board

STM32 NUCLEO-G491RE PIN MAPPING:

SIGNAL	PIN	TIMER / AF	CONNECTOR
X_STEP	PA8	TIM1_CH1 / AF6	CN10 pin 23 (D7)
X_DIR	PB0	GPIO Output	CN7 pin 34 (A3)
Y_STEP	PA9	TIM1_CH2 / AF6	CN10 pin 21 (D8)
Y_DIR	PB1	GPIO Output	CN10 pin 24
Z_STEP	PA5	TIM2_CH1 / AF1	CN10 pin 11 (D13)
Z_DIR	PB2	GPIO Output	CN10 pin 22
GND	GND	Digital Ground	CN10 pin 20 / 9

6-CH NMOS LEVEL SHIFTER SINK BOARD:

CH	GATE INPUT	DRAIN SINK OUTPUT
CH 1	STM32 PA8 (X_STEP)	Sinks Driver X PUL-
CH 2	STM32 PB0 (X_DIR)	Sinks Driver X DIR-
CH 3	STM32 PA9 (Y_STEP)	Sinks Driver Y PUL-
CH 4	STM32 PB1 (Y_DIR)	Sinks Driver Y DIR-
CH 5	STM32 PA5 (Z_STEP)	Sinks Driver Z PUL-
CH 6	STM32 PB2 (Z_DIR)	Sinks Driver Z DIR-
ANODE	+24VDC (V-PULSE)	Feeds all PUL+ & DIR+
SOURCE	MCU GND + DC 0V	Common Return Reference

CRITICAL HARDWARE WIRING WARNINGS:

- DO NOT plug PB0 into CN10-31! (On Nucleo-64 that is PB3). Connect to CN7 pin 34.
- DO NOT plug PB1 into CN10-7! (That is AVDD 3.3V). Connect to CN10 pin 24.
- DO NOT plug PB2 into CN10-15! (That is PA7). Connect to CN10 pin 22.
- DO NOT connect GND to CN10-5, 17, 27! Use CN10 pin 20 or CN10 pin 9 only.
- PA5 shares onboard LD2 LED: Firmware MUST disable all LED blink functions.
- Resistors: Rg = 100 Ohm (damping), Rpd = 10k Ohm (tri-state pulldown).

Failure to observe these pin rules will lock direction or damage MCU pins.